

Final Mini-Project

*Does Poverty Drive Drug Overdose ED Visits
in Virginia Counties?*

Virginia counties with higher poverty rates experience higher drug overdose ED visit rates.

Rationale

- Poverty is associated with limited access to healthcare, mental health resources, and social support — all risk factors for substance use disorder.
- If the proposition holds, we expect average overdose ED visit rates to rise monotonically across Low → Medium → High poverty buckets.
- Educational attainment is used as a corroborating lens: counties with more bachelor's degree holders should show lower overdose rates.

DATASETS

VDH Overdose ED Visits

Virginia Department of Health / VA-ODP

Drug overdose ED visit counts and rates per 10,000 visits by locality, year, and drug type. Filtered to: All Drug, Locality level, Year 2023, non-combined localities.

89 records after cleaning.

ACS S1702 — Poverty Status

US Census Bureau, 2023 5-Year Estimates

Poverty rate (% families below poverty level) and educational attainment by county across all US counties. Filtered to Virginia counties only. 133 Virginia counties.

After joining on county name: 74 matched Virginia localities

METHODOLOGY

1

Load & Clean

Loaded both CSVs using Spark DataFrameReader. Filtered overdose data to All Drug / Locality / 2023 / non-combined. Dropped nulls.

2

Join

Inner joined on county name (standardized by stripping ', Virginia' from ACS names). Result: 74 matched counties.

3

Bucket by Poverty

Classified counties into Low (<8%), Medium (8–15%), High (>15%) poverty buckets. Computed average OD rate per bucket.

4

Bucket by Education

Derived % bachelor's degree holders per county. Bucketed into Low/Medium/High. Computed average OD rate per bucket.

5

Correlation

Computed Pearson correlation between poverty rate and OD rate using Spark's DataFrame.stat.corr().

DATA SAMPLES — LOADED IN SPARK

Overdose ED Visits (top 5 rows)

```
[scala> :load miniproject_spark.scala
val args: Array[String] = Array()
Loading miniproject_spark.scala...
import org.apache.spark.sql.functions._
val overdoseRaw: org.apache.spark.sql.DataFrame = [Data Extract Date: string, Overdose ED Visit Year: int ... 7 more fields]
val overdose: org.apache.spark.sql.DataFrame = [locality: string, fips: string ... 1 more field]
=== Overdose Data Sample ===
+-----+-----+-----+
|locality|fips|od_rate|
+-----+-----+-----+
|Middlesex|51119|28.5|
|Scott|51169|42.0|
|Lancaster|51103|48.4|
|Craig|51045|48.4|
|Pulaski|51155|70.2|
+-----+-----+-----+
only showing top 5 rows
Overdose record count: 89
val acsRaw: org.apache.spark.sql.DataFrame = [GEO_ID: string, NAME: string ... 601 more fields]
val acsClean: org.apache.spark.sql.DataFrame = [county: string, pct_poverty: double ... 3 more fields]
```

ACS Poverty Data (top 5 rows)

```
=== ACS Data Sample ===
+-----+-----+-----+-----+-----+
|county|pct_poverty|count_less_than_hs|count_bachelors_plus|total_families|
+-----+-----+-----+-----+-----+
|Accomack|10.2|1349.0|2452.0|9145.0|
|Albemarle|3.7|1281.0|18605.0|28029.0|
|Alleghany|9.7|239.0|785.0|3759.0|
|Amelia|4.9|389.0|735.0|3616.0|
|Amherst|7.3|705.0|2166.0|8134.0|
+-----+-----+-----+-----+-----+
only showing top 5 rows
ACS record count: 133
val joined: org.apache.spark.sql.DataFrame = [locality: string, od_rate: double ... 4 more fields]
```

JOINED DATASET (74 COUNTIES)

Inner join on county name — overdose data ✕ ACS poverty data

```
=== Joined Dataset ===
```

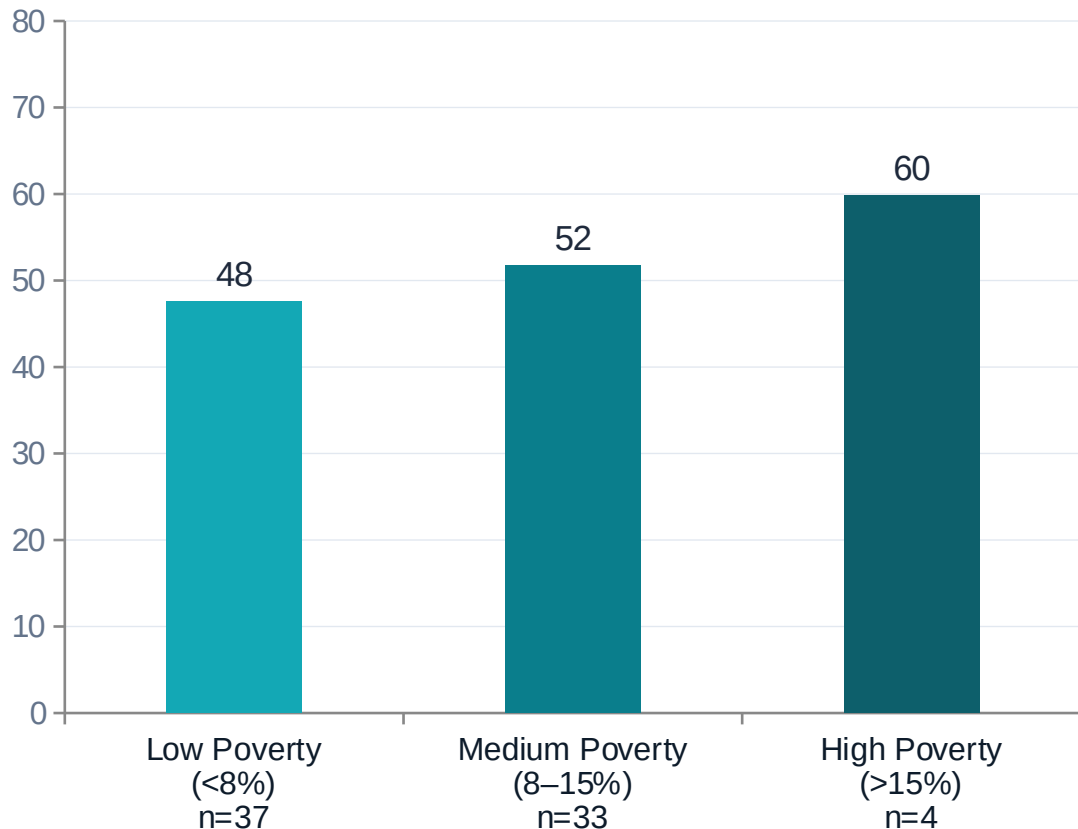
locality	od_rate	pct_poverty	count_less_than_hs	count_bachelors_plus	total_families
Middlesex	28.5	8.3	124.0	1158.0	3012.0
Scott	42.0	13.6	833.0	879.0	5451.0
Lancaster	48.4	10.4	262.0	1409.0	3186.0
Craig	48.4	10.4	74.0	222.0	1304.0
Pulaski	70.2	8.8	504.0	2484.0	8850.0
King William	39.5	5.9	263.0	1590.0	5218.0
Prince Edward	43.1	11.2	284.0	1398.0	4434.0
Stafford	39.4	3.6	2227.0	21020.0	39989.0
Smyth	57.3	11.6	768.0	1487.0	7903.0
Buckingham	72.9	11.0	563.0	555.0	4031.0

only showing top 10 rows
Joined record count: 74

```
val bucketed: org.apache.spark.sql.DataFrame = [locality: string, od_rate: double ... 5 more fields]  
val avgByPoverty: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [poverty_bucket: string, avg_od_rate: double ... 2 more fields]
```

74 out of 133 Virginia counties matched successfully. Unmatched counties are due to name formatting differences (e.g. combined localities).

FINDING 1 — POVERTY vs. OVERDOSE RATE



Spark Output

```
== Avg OD Rate by Poverty Bucket ==  
+-----+-----+-----+-----+  
| poverty_bucket | avg_od_rate | num_counties | avg_pct_poverty |  
+-----+-----+-----+-----+  
| 1_Low (<8%)   | 47.57567567567569 | 37           | 4.832432432432434 |  
| 2_Medium (8-15%) | 51.727272727272734 | 33           | 10.554545454545455 |  
| 3_High (>15%)  | 59.775000000000006 | 4           | 19.475           |  
+-----+-----+-----+-----+  
  
val withEduPct: org.apache.spark.sql.DataFrame = [locality: string, od_rate: double ... 5 more fields]  
val eduBucketed: org.apache.spark.sql.DataFrame = [locality: string, od_rate: double ... 6 more fields]  
val avgByEdu: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [edu_bucket: string, avg_od_rate: double ... 2 more fields]
```

✓ Clear upward trend:

Low poverty counties average 47.6 vs High poverty at 59.8 — a 26% gap.

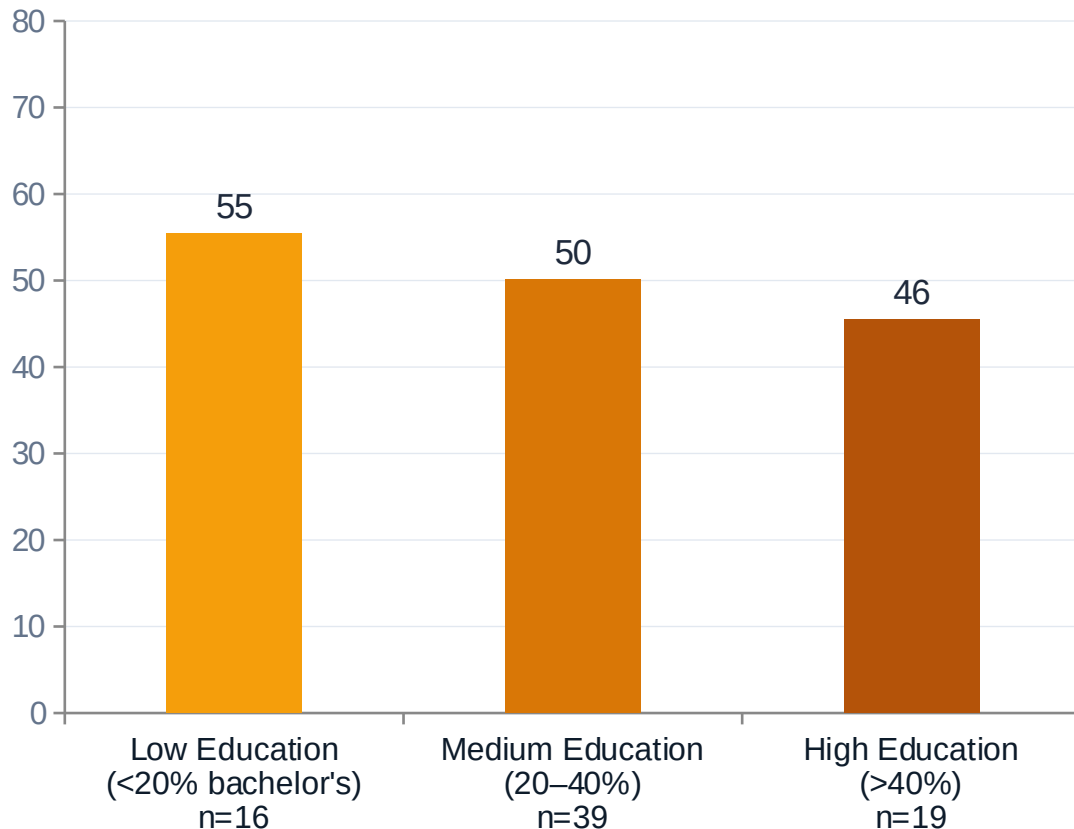
✓ Supports the proposition:

Each bucket is higher than the last.

⚠ Note:

High poverty bucket has only 4 counties — interpret with caution.

FINDING 2 — EDUCATION vs. OVERDOSE RATE



Spark Output

```
=== Avg OD Rate by Education Bucket ===
```

edu_bucket	avg_od_rate	num_counties	avg_pct_bachelors
1_Low (<20% bachelors)	55.43749999999999	16	16.95328274124196
2_Medium (20-40%)	50.12051282051283	39	28.471729868611757
3_High (>40%)	45.51052631578948	19	50.0452846944692

✓ Clear downward trend:

Low education counties average 55.4 vs High education at 45.5 — an 18% gap.

✓ Corroborates Finding 1:

Poverty and low education tend to co-occur — both independently predict higher OD rates.

✓ Consistent direction:

Each education bucket is lower than the last.

TOP & BOTTOM COUNTIES BY OD RATE

Top 10 Highest OD Rate Counties

=== Top 10 Counties by OD Rate ===

locality	od_rate	pct_poverty	count_less_than_hs	count_bachelors_plus	total_families
Amherst	121.9	7.3	705.0	2166.0	8134.0
Carroll	107.3	11.2	964.0	1618.0	8097.0
Appomattox	101.2	8.7	320.0	1046.0	4535.0
Patrick	95.1	8.1	874.0	854.0	4927.0
Buchanan	91.7	17.9	1181.0	703.0	4956.0
Bland	75.5	8.6	62.0	299.0	1485.0
Henrico	74.9	6.0	4329.0	44080.0	85753.0
Sussex	73.8	6.9	277.0	525.0	2400.0
Bedford	73.0	5.1	1272.0	8923.0	22887.0
Buckingham	72.9	11.0	563.0	555.0	4031.0

only showing top 10 rows

Bottom 10 Lowest OD Rate Counties

=== Bottom 10 Counties by OD Rate ===

locality	od_rate	pct_poverty	count_less_than_hs	count_bachelors_plus	total_families
Highland	0.0	3.6	82.0	251.0	645.0
Charles City	17.3	8.2	162.0	349.0	1787.0
Northumberland	20.6	5.6	259.0	1376.0	3433.0
Rappahannock	21.4	4.5	160.0	988.0	1995.0
Westmoreland	24.0	9.1	689.0	1154.0	4580.0
Floyd	25.9	4.4	216.0	1021.0	4286.0
Dinwiddie	27.0	8.4	361.0	1903.0	7427.0
Middlesex	28.5	8.3	124.0	1158.0	3012.0
Lunenburg	28.9	13.6	580.0	415.0	3008.0
Amelia	29.3	4.9	389.0	735.0	3616.0

only showing top 10 rows

val correlation: Double = 0.17175785236346622

=== Correlation (poverty vs OD rate): 0.17175785236346622 ===

Notable: Amherst (121.9 rate, 7.3% poverty), Carroll (107.3, 11.2%), Buchanan (91.7, 17.9% — highest poverty in top 10). Low-rate counties like Highland and Rappahannock tend to have lower poverty.

CORRELATION ANALYSIS

Pearson Correlation

$$r = 0.172$$

poverty rate vs. OD rate (n=74 counties)

```
only showing top 10 rows  
val correlation: Double = 0.17175785236346622  
== Correlation (poverty vs OD rate): 0.17175785236346622 ==
```

Weak positive correlation ($r = 0.172$):

Poverty rate is positively associated with OD rate, but the relationship is not strong on its own.

Why weak?

OD rates are driven by many factors beyond poverty — prescription pill access, local drug markets, rural isolation. These confounders dilute the signal.

CONCLUSION

Evidence partially supports the proposition.

Higher poverty counties do show higher average OD rates across all three buckets.

✓ **Poverty bucket analysis**

Low (47.6) → Medium (51.7) → High (59.8) — consistent upward trend.

✓ **Education bucket analysis**

Higher education = lower OD rate, corroborating the poverty finding.

⚠ **Correlation ($r = 0.172$) is weak**

Poverty is one of many drivers. Rural access, local drug markets, and healthcare availability also play major roles.

AI USAGE — PROMPTS USED

Claude (Anthropic) was used to assist in converting analysis ideas into working Spark Scala code.

Prompt 1 — Data Cleaning

Using the county column in the ACS S1702 data and the locality column in the VDH overdose data, write Spark Scala to clean both datasets and standardize county names for joining.

Prompt 2 — Join Logic

Using the cleaned ACS and overdose DataFrames, write Spark SQL to inner join on county name, keeping od_rate, pct_poverty, and education count columns.

Prompt 3 — Bucketing & Aggregation

Using the joined Spark DataFrame, write Scala to bucket counties by poverty rate into Low/Medium/High and compute the average OD rate per bucket using groupBy and agg.

Prompt 4 — Correlation

Using the joined Spark DataFrame with pct_poverty and od_rate columns, write Scala to compute the Pearson correlation coefficient.